NOT MEASUREMENT SENSITIVE

MIL-P-85856/12
7 December 1988

MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS NICKEL BASE ALLOY, TYPE I, CLASS 1, COMPOSITION K

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Mo	Cr	Ni	Fe	V	C	Mn	Si	Co	P	S
23.0 26.0	4.0 6.0	Bal	4.0 7.0	0.6 max.	0.12 max.	1.0 max.	1.0 max.	2.5 max.	0.02 max.	0.02 max.

Particle Size/Distribution

-230 Mesh (-63 µm)	95.0 min.
-270 Mesh (-53 µm)	80.0 min.

Sprayed Coating Properties

Bond Strength	31.0 MPa (4500 psi) min.
Hardness	Rockwell 15T, 90 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 3% max. Unmelted Particles - 10% max.

AMSC N/A FSC 3439

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Deposit Efficiency	70% min for N_2/H_2 65% min for A/H_2
Deposition Rate _____	lb/hr
_____	g/min

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/12)

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